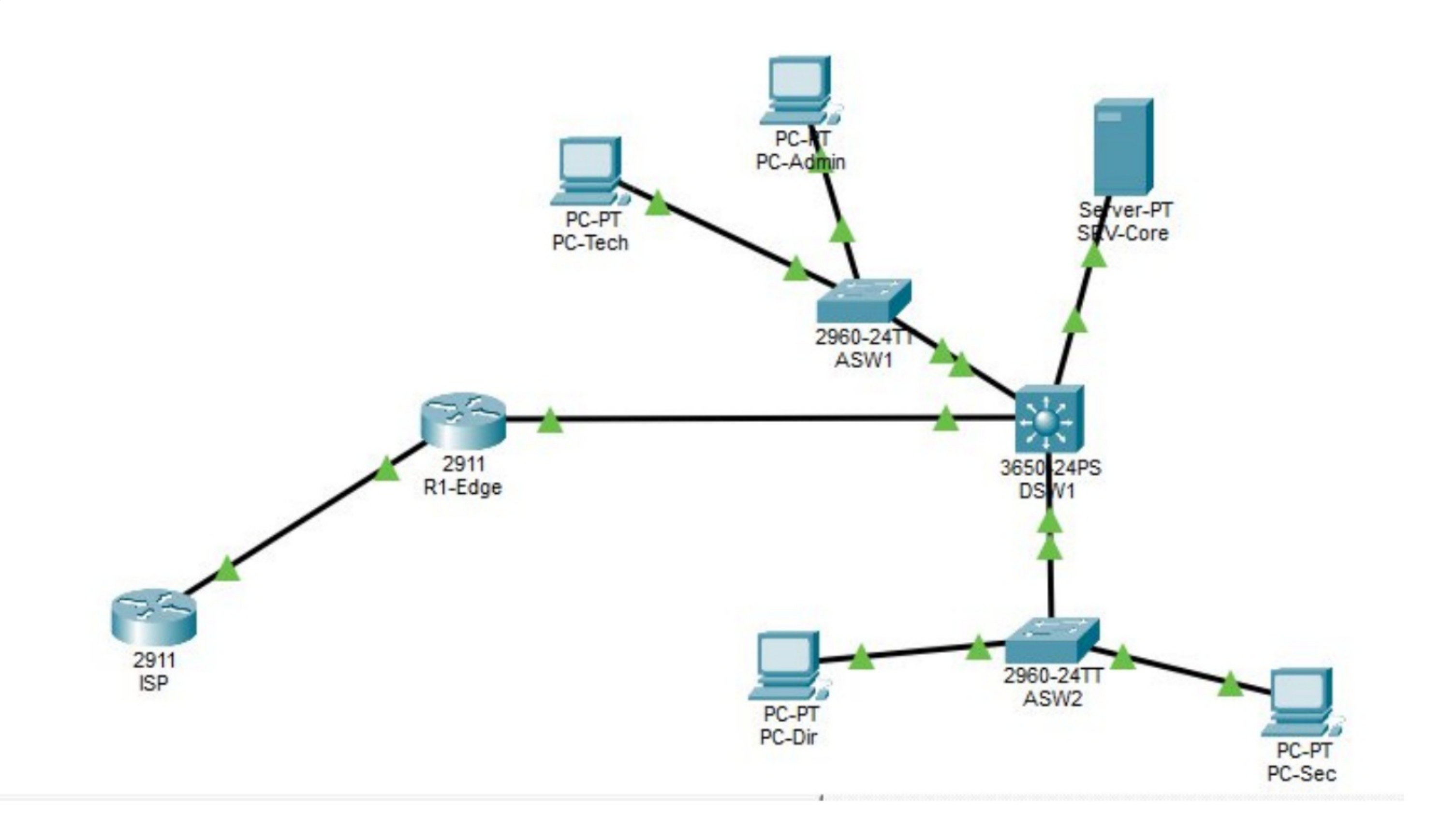
Enterprise Multi-Layer Network Security

Scope: Cisco IOS network engineering procedure — multi-layer architecture validation **Target:** Cisco CCNA 200-301 exam & enterprise deployment



Part 1 — Core Concepts

1. Multi-Layer Switching (Layer 3) and SVIs

In a modern enterprise network architecture, inter-VLAN routing is centralized on a Layer 3 switch (Multilayer Switch), such as the Cisco Catalyst 3650. Unlike traditional Router-on-a-Stick (RoAS) routing, which can create bandwidth bottlenecks on a single link, a Layer 3 distribution switch routes packets at hardware speed (ASIC). This relies on **Switch Virtual Interfaces (SVIs)** — each SVI acts as the default gateway for its respective VLAN segment, providing high-performance internal routing without involving the edge router.

2. Routed Ports vs. Switchports

To interconnect the network core (DSW1) with the edge router (R1-EDGE), the architecture bypasses Layer 2 mechanisms (Spanning Tree, VLANs) on this specific link. Applying the `no switchport` command on a Layer 3 switch's physical interface converts it into a **routed port**. It then behaves strictly as a pure router interface, accepting a direct point-to-point IP address (/30 mask) and running routing protocols — isolating the LAN broadcast domain from the edge.

3. PAT (Port Address Translation) and Perimeter Filtering

Under RFC 1918, private IP address blocks used inside the internal infrastructure (10.10.0.0/16) cannot be routed on the public Internet. To provide outbound connectivity, the R1-EDGE router implements **Dynamic NAT with Overload (PAT)**. This maps thousands of simultaneous private sessions to a single routable public IP address by dynamically rewriting the TCP/UDP source ports in the Layer 4 headers. Security is reinforced by the router's strict state table, which blocks any unsolicited inbound connection attempt.

4. DHCP Relay (`ip helper-address`)

Initial DHCP discovery messages (DHCP DISCOVER) are sent as Layer 2/Layer 3 broadcasts. By design, routed interfaces (including DSW1's SVIs) block these broadcasts to avoid flooding network segments. With the DHCP server (SRV-CORE) centralized in a dedicated VLAN (VLAN 30), client requests from VLAN 10 and VLAN 20 are intercepted by the `ip helper-address` command: the switch encapsulates the local broadcast into a targeted unicast packet, forwarded through the routing table directly to the DHCP server's IP address.

Part 2 — Topology & Addressing Matrix

1. Physical Architecture & Interconnections

- **IT segment (VLAN 10):** hosted on switch ASW1 (FastEthernet 0/1 & 0/2)
- **Management segment (VLAN 20):** hosted on switch ASW2 (FastEthernet 0/1 & 0/2)
- **Uplink trunks:** redundant Gigabit Ethernet links from ASW1 (Gi0/1) and ASW2 (Gi0/1) to the core switch DSW1's Gi1/0/1 and Gi1/0/2
- **Server / internal DMZ (VLAN 30):** infrastructure server SRV-CORE connected on DSW1's dedicated port Fa1/0/24
- **Core-to-edge interconnection:** dedicated point-to-point link via a routed port between DSW1 (Gi1/0/3) and R1-EDGE (Gi0/0)
- **Simulated WAN link:** external public segment connecting R1-EDGE (Gi0/1) to the ISP router (Gi0/0)

2. IP Addressing & VLAN Segmentation

Device / Interface	Role / VLAN	IP Network / Prefix	Fixed IP Address	Gateway / Next Hop
DSW1 – SVI VLAN 10	IT gateway	10.10.10.0/24	10.10.10.254	N/A
DSW1 – SVI VLAN 20	Management gateway	10.10.20.0/24	10.10.20.254	N/A
DSW1 – SVI VLAN 30	Servers gateway	10.10.30.0/24	10.10.30.254	N/A
DSW1 – Gi1/0/3	Routed port (to edge)	10.10.100.0/30	10.10.100.2	N/A
R1-EDGE – Gi0/0	Internal LAN interface	10.10.100.0/30	10.10.100.1	N/A
R1-EDGE – Gi0/1	Public WAN interface	203.0.113.0/30	203.0.113.1	N/A
ISP – Gi0/0	ISP gateway	203.0.113.0/30	203.0.113.2	N/A
ISP – Loopback 0	Simulated Internet host	8.8.8.8/32	8.8.8.8	N/A
SRV-CORE	DHCP / DNS server	VLAN 30	10.10.30.1	10.10.30.254
PC-Admin / PC-Tech	IT workstations	VLAN 10	Dynamic (DHCP)	10.10.10.254
PC-Dir / PC-Sec	Management workstations	VLAN 20	Dynamic (DHCP)	10.10.20.254

Part 3 — Cisco IOS Configuration Scripts

1. Access Switch ASW1 (Layer 2)

```
ASW1> enable
ASW1# configure terminal
ASW1(config)# hostname ASW1
! --- CREATE ACCESS VLAN ---
ASW1(config)# vlan 10
ASW1(config-vlan)# name Informatique
ASW1(config-vlan)# exit
! --- ASSIGN HOST PORTS ---
ASW1(config)# interface range fastEthernet 0/1 - 2
ASW1(config-if-range)# switchport mode access
ASW1(config-if-range)# switchport access vlan 10
ASW1(config-if-range)# exit
! --- CONFIGURE UPLINK TRUNK ---
ASW1(config)# interface gigabitEthernet 0/1
ASW1(config-if)# switchport mode trunk
ASW1(config-if)# exit
ASW1(config)# end
ASW1# copy running-config startup-config
```

2. Access Switch ASW2 (Layer 2)

```
ASW2> enable
ASW2# configure terminal
ASW2(config)# hostname ASW2
! --- CREATE ACCESS VLAN ---
ASW2(config)# vlan 20
ASW2(config-vlan)# name Direction
ASW2(config-vlan)# exit
! --- ASSIGN HOST PORTS ---
ASW2(config)# interface range fastEthernet 0/1 - 2
ASW2(config-if-range)# switchport mode access
ASW2(config-if-range)# switchport access vlan 20
ASW2(config-if-range)# exit
! --- CONFIGURE UPLINK TRUNK ---
ASW2(config)# interface gigabitEthernet 0/1
ASW2(config-if)# switchport mode trunk
ASW2(config-if)# exit
ASW2(config)# end
ASW2# copy running-config startup-config
```

3. Multi-Layer Core Switch (DSW1) — Full Configuration

```
DSW1> enable
DSW1# configure terminal
DSW1(config)# hostname DSW1
! --- INITIALIZE VLAN DATABASE ---
DSW1(config)# vlan 10
DSW1(config-vlan)# name Informatique
DSW1(config)# vlan 20
DSW1(config-vlan)# name Direction
DSW1(config)# vlan 30
DSW1(config-vlan)# name Serveurs
DSW1(config-vlan)# exit
! --- ENABLE TRUNK ON MODERN CATALYST (NATIVE 802.1Q) ---
DSW1(config)# interface range gigabitEthernet 1/0/1 - 2
DSW1(config-if-range)# switchport mode trunk
DSW1(config-if-range)# exit
! --- DEPLOY SERVER PORT INTO ITS VLAN ---
DSW1(config)# interface gigabitEthernet 1/0/24
DSW1(config-if)# switchport mode access
DSW1(config-if)# switchport access vlan 30
DSW1(config-if)# exit
! --- LAYER 3 SWITCHING: ENABLE SYSTEM ROUTING ENGINE ---
DSW1(config)# ip routing
! --- SVI CONFIGURATION & DHCP RELAY ---
DSW1(config)# interface vlan 10
DSW1(config-if)# ip address 10.10.10.254 255.255.255.0
DSW1(config-if)# ip helper-address 10.10.30.1
DSW1(config-if)# no shutdown
DSW1(config-if)# exit
DSW1(config)# interface vlan 20
DSW1(config-if)# ip address 10.10.20.254 255.255.255.0
DSW1(config-if)# ip helper-address 10.10.30.1
DSW1(config-if)# no shutdown
DSW1(config-if)# exit
DSW1(config)# interface vlan 30
DSW1(config-if)# ip address 10.10.30.254 255.255.255.0
DSW1(config-if)# no shutdown
DSW1(config-if)# exit
! --- CRITICAL STEP: CREATE THE ROUTED PORT TO THE EDGE ---
DSW1(config)# interface gigabitEthernet 1/0/3
DSW1(config-if)# no switchport
DSW1(config-if)# ip address 10.10.100.2 255.255.255.252
DSW1(config-if)# no shutdown
DSW1(config-if)# exit
! --- DEFAULT ROUTE (GATEWAY OF LAST RESORT) ---
DSW1(config)# ip route 0.0.0.0 0.0.0.0 10.10.100.1
DSW1(config)# end
DSW1# copy running-config startup-config
```

4. Security Edge Router (R1-EDGE)

```
R1-Edge> enable
R1-Edge# configure terminal
R1-Edge(config)# hostname R1-Edge
! --- LAN INTERCONNECTION LINK ---
R1-Edge(config)# interface gigabitEthernet 0/0
R1-Edge(config-if)# ip address 10.10.100.1 255.255.255.252
R1-Edge(config-if)# ip nat inside
R1-Edge(config-if)# no shutdown
R1-Edge(config-if)# exit
! --- WAN INTERNET UPLINK ---
R1-Edge(config)# interface gigabitEthernet 0/1
R1-Edge(config-if)# ip address 203.0.113.1 255.255.255.252
R1-Edge(config-if)# ip nat outside
R1-Edge(config-if)# no shutdown
R1-Edge(config-if)# exit
! --- PERIMETER STATIC ROUTING ---
R1-Edge(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.2
R1-Edge(config)# ip route 10.10.0.0 255.255.0.0 10.10.100.2
! --- NAT/PAT OVERLOAD MECHANISM ---
R1-Edge(config)# access-list 1 permit 10.10.0.0 0.0.255.255
R1-Edge(config)# ip nat inside source list 1 interface gigabitEthernet 0/1 overload
R1-Edge(config)# end
R1-Edge# copy running-config startup-config
```

5. External Operator Router (ISP)

```
Router> enable
Router# configure terminal
Router(config)# hostname ISP
! --- CUSTOMER LINK SEGMENT ---
ISP(config)# interface gigabitEthernet 0/0
ISP(config-if)# ip address 203.0.113.2 255.255.255.252
ISP(config-if)# no shutdown
ISP(config-if)# exit
! --- LOOPBACK INTERFACE SIMULATING THE INTERNET ---
ISP(config)# interface loopback 0
ISP(config-if)# ip address 8.8.8.8 255.255.255.255
ISP(config-if)# exit
ISP(config)# end
ISP# copy running-config startup-config
```

Part 4 — Validation, Traffic Audit & Incident Log

1. End-to-End Test Plan

- **Dynamic addressing layer:** open the network configuration panel on PC-Admin and PC-Tech. Confirm DHCP provides a valid configuration in the 10.10.10.X range, with gateway 10.10.10.254 and DNS server 10.10.30.1.
- **Local inter-VLAN routing audit:** run `ping 10.10.30.1` from PC-Admin's terminal. Success rate must be 100%, confirming DSW1 correctly switches traffic between VLAN 10 and VLAN 30.
- **Global WAN connectivity test:** from PC-Admin's command prompt, run `ping 8.8.8.8`. The first packet may time out during ARP discovery; subsequent packets must receive an explicit reply from the ISP's virtual interface.

2. IOS Inspection Commands

```
DSW1# show ip route
! Confirms local access segments and the gateway of last resort (0.0.0.0/0 via 10.10.100.1)

R1-Edge# show ip nat translations
! Critical audit command: shows in real time the private IP of the internal host (Inside Local)
! dynamically translated to the public WAN IP (Inside Global), with strict Layer 4 port mapping
```

🛡 **Security audit note:** running `show ip nat translations` on the edge router confirms the LAN is fully opaque from outside. If an attacker on the Internet side (ISP) sniffs traffic, the original source IP headers of internal hosts (e.g. `10.10.10.10`) remain completely invisible, masked behind the legitimate public address `203.0.113.1`.

3. Incident Log

Incident #1 — Trunk encapsulation command rejected

- **Symptom:** `Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "Trunk" mode.`
- **Root cause:** hardware model mismatch. On older Cisco switch generations (e.g. Catalyst 3560), the administrator had to manually specify the encapsulation algorithm (Cisco's proprietary ISL or the IEEE 802.1Q standard). Modern core equipment like the Catalyst 3650 natively defaults to 802.1Q and has fully dropped legacy ISL support, so the CLI rejects the now-obsolete command.
- **Fix:** remove the redundant encapsulation line and force the port directly into operational trunk mode with `switchport mode trunk` alone.

Incident #2 — DHCP server pool conflict in Packet Tracer

- **Symptom:** client requests fail with a critical red warning: `DHCP request failed`. Analysis shows the default pool overrides the manually written configuration.
- **Root cause:** an internal quirk of Cisco Packet Tracer's simulation engine. The generic DHCP service ships with a persistent pool named `serverPool` that auto-activates and takes precedence when the system engine loads. If the operator also configures a manual duplicate named `VLAN10` while leaving the global engine enabled, the simulator creates a logical conflict in IP lease distribution, breaking delivery to the LAN.
- **Fix:** select the default `serverPool` entry, inject the strict VLAN 10 parameters into it (gateway `10.10.10.254`, DNS `10.10.30.1`), then click **Save** to overwrite the block. Remove the failing manual duplicate with the **Remove** button. Finally, set the global services toggle to **On** to stabilize distribution.

Incident #3 — Hosts isolated in the APIPA address block

- **Symptom:** PCs on the infrastructure can no longer reach their default gateway. `ipconfig` shows an abnormal IP address in the `169.254.184.101` subnet.
- **Root cause:** APIPA (Automatic Private IP Addressing). During infrastructure startup (particularly after power-cycling the modular power supplies on DSW1), switches run the Spanning Tree Protocol (STP) algorithm to rule out loops. This computation phase keeps ports blocked for 30–50 seconds. Client PCs send DHCP requests during this window and, receiving no reply due to the temporary block, their NICs fall back to the APIPA emergency block (`169.254.0.0/16`).
- **Fix:** reset the client network stack once the infrastructure has stabilized (link LEDs solid green). In Packet Tracer's GUI, toggle the NIC to **Static** and immediately back to **DHCP**. Alternatively, run `ipconfig /release` followed by `ipconfig /renew` at the system prompt.